Kriti Sehgal

Preceptor in Data Science, Data Science Institute, The University of Chicago
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Research and Teaching Interests: Data Science, Mathematical Analysis, Data Science Education, Mathematics Education.

EDUCATION

The Ohio State University (OSU) — Columbus, OH, USA
PhD, Mathematics (advisor: Prof. Ovidiu Costin) — *Aug 2018 – Apr 2024*
MS, Mathematics — *Aug 2018 – Jul 2021*

Indian Institute of Science (IISc) — Bangalore, India
MS, Mathematics (advisor: Prof. Tirthankar Bhattacharyya) — *Jul 2015 – Jun 2018*

University of Delhi (DU) — Delhi, India
BSc (Honors), Mathematics — *Jul 2012 – May 2015*

WORK EXPERIENCE

Data Science Institute, University of Chicago — Chicago, IL, USA
Preceptor in Data Science — *Jul 2024 – Present*

Wolfram|Alpha — Champaign, IL, USA
Scientific Content Development Intern — *Jun 2023 – Aug 2023*

TEACHING EXPERIENCE

Syllabi and student evaluations: kritisehgal01.github.io

University of Chicago, *Instructor*

- DATA 211 – Mathematical Methods for Data Science I — Fall 2026
- DATA 231 – Machine Learning Fundamentals: Theory and Practice — Fall 2026, Spring 2026
- DATA 10018 – Pathways in Data Science — Summer 2026
- DATA 213 – Models in Data Science, Co-instructor — Spring 2026
- DATA 119 – Introduction to Data Science II — Winter 2026
- DATA 119 Labs – Introduction to Data Science II — Spring 2026, Fall 2025
- DATA 118 – Introduction to Data Science I — Spring 2025

City Colleges of Chicago (CCC), *Instructor*

- CIS 175 – Introduction to Data Science, Olive-Harvey College — Fall 2026
- MATH 118 – General Education Math, Kennedy-King College — Spring 2026
- MATH 125 – Introduction to Statistics, Olive-Harvey College — Spring 2025

The Ohio State University (OSU), *Graduate Teaching Associate*

- MATH 1149 – Trigonometry — Spring 2024
- MATH 2173 – Calculus III (Engineering Mathematics B) — Fall 2023
- MATH 1131 – Calculus for Business — Spring 2022
- MATH 1156 – Calculus for Biological Sciences — Fall 2021
- MATH 2153 – Calculus III — Spring 2021
- MATH 1172 – Calculus II (Engineering Mathematics) — Spring 2020, Fall 2020
- MATH 1151 – Calculus I — Fall 2019

Indian Institute of Science (IISc), *Teaching Assistant*

- UM 101 – Undergraduate Analysis and Linear Algebra — Fall 2017

CURRICULUM DEVELOPMENT

Courses and Programs

- **Community College Partnership and Course Development:** Developed and currently teach the first data science curriculum for high school students enrolled in a college-level programme at Olive-Harvey College (CCC).
- **Pathways in Data Science Curriculum Development (Summer 2026):** Developed the curriculum for a summer immersive data science program at UChicago for advanced high school students from around the world.
- **Lab Curriculum Review and Facilitation:** Reviewed and refined DATA 119 lab materials as part of a teaching research initiative investigating the impact of mandatory labs on student learning in data science.

Book Chapters

1. London, J. and Sehgal, K. **DataFrames**. In *The Chicago Guide to Data Science*. [Link]
2. Sehgal, K. **Introduction to Machine Learning**. In *The Chicago Guide to Data Science*. [Link]
3. Sehgal, K. **Neural Networks and Deep Learning**. In *The Chicago Guide to Data Science*. [Link]
4. Sehgal, K. **Tree Models**. In *The Chicago Guide to Data Science*. [Link]

MENTORSHIP

University of Chicago

- Data Science Clinic program — Fall 2026, Winter 2026, Spring 2025, Winter 2025, Fall 2024
- Data Science for Social Impact Summer Experience — Summer 2025

The Ohio State University (OSU)

- Directed reading program — Fall 2021
- One-on-one mentorship to train new teaching associates — Fall 2021

AWARDS AND ACHIEVEMENTS

- **Teaching**
  - 2023 Graduate Associate Teaching Award (GATA), the highest teaching honor for graduate teaching associates at OSU.
  - 2022 Phil Huneke Distinguished Graduate Teaching Associate Award, Department of Mathematics, OSU.
- **Service and Leadership**
  - 2023 SIAM Student Chapter Certificate of Recognition acknowledging exceptional service to OSU's SIAM Chapter.
  - Nominated for 2021 Graduate Student Leadership Award, OSU's highest recognition for student leadership.
- **Academic Honors**
  - Qualified National Eligibility Test for Lectureship (NET), India in Mathematics.
  - All India Rank 5 in Joint Admission Test for M.Sc. (JAM), Mathematics, 2015.
  - First Position in undergraduate studies at University of Delhi.

RESEARCH PROJECTS

Publications and Theses

1. Long time evolution of the Hénon-Heiles system for small energy, *Journal of Mathematical Physics 66, no. 9, 2025*, O. Costin, R. Costin, K. Sehgal.
2. On the pointwise existence of Cauchy P.V. integrals, (*Under Review*, arXiv), N. Castillo, O. Costin, K. Sehgal.
3. PhD thesis: Dynamics of the Hénon-Heiles System and Generalizing the Sokhotski-Plemelj Formula.
4. M.S. thesis: Duality for Spaces of Holomorphic Functions into a Locally Convex Topological Vector Space.

Selected Projects in Data Science Workshops

1. Data Science for Social Impact Summer 2025: Supervised a project developing *Seedbot*, a chatbot that parses legal documents and answers questions about seed laws across 78 countries in simplified language; *poster published*.
2. Mathematical Problems in Industry (MPI 2025): Collaborated with Kwaai, an open-source AI lab, to study privacy-preserving query methods for vector databases in Personal AI. Explored dimensional scrambling, noise injection, and cryptographic methods such as ElGamal and CKKS; contributed to the development of homomorphic encryption algorithms.
3. Mathematical Problems in Industry (MPI 2024): Analyzed Vironix Health's de-identified remote patient monitoring data to identify positive health outcomes and predict adverse health episodes, patient compliance, and engagement.
4. Wolfram|Alpha: Contributed to a project extracting and validating mathematical assertions from scientific papers, using LaTeX parsing, pattern matching, and regular expressions.

LEADERSHIP AND SERVICE

- Teaching Assistant and mentor, Statistics for Machine Learning, Data Science, and Artificial Intelligence training program for undergraduates by MTTS, India — Oct 11 – Nov 8, 2025
- President, Society for Industrial and Applied Mathematics (SIAM) Student Chapter, OSU — Aug 2022 – Apr 2023
- Vice President, Association for Women in Mathematics (AWM) Student Chapter, OSU — Aug 2021 – Apr 2023
- Vice President, Mathematics Graduate Student Association, OSU — Aug 2020 – May 2022
- Founder and Organizer, Student Analysis Seminar, Department of Mathematics, OSU — Jan 2020 – May 2022
- Graduate Student Representative, Department of Mathematics Diversity Committee, OSU — Aug 2020 – May 2021
- Outreach Coordinator, SIAM Student Chapter, IISc — Aug 2017 – May 2018

TALKS

- Invited Speaker, Data4All, University of Chicago, Chicago, IL — Nov 23, 2024
- Invited Speaker, SIAM Great Lakes Conference, East Lansing, MI — Oct 14, 2023
- AWM Poster Presentation, Joint Mathematics Meetings — Jan 8, 2021

PROFESSIONAL DEVELOPMENT

Trainings

- **Faculty Development Programme, City Colleges of Chicago (CCC), Fall 2024 – Spring 2026:** Completed faculty development training in evidence-based pedagogy, inclusive teaching, classroom action research, and reflective practice.

Selected Conferences, Workshops, and Summer Schools

1. National Workshop on Data Science Education: *June 22–24, 2026; June 24–27, 2025*.
2. Mathematical Problems in Industry (MPI): *June 14–18, 2026; June 9–13, 2025; June 25–29, 2024; June 13–17, 2022*.
3. SIAM Workshop "From Machine Learning to Large Language Models – An Introduction": *October 20, 2024*.
4. Teaching in the Generative AI Landscape at University of Chicago: *September 4–5, 2024*.
5. The Erdős Institute Data Science Boot Camp: *Aug–Dec, 2023; May 2022*.
6. Graduate Student Math Modeling Camp 2022: *June 8–11, 2022*.
7. Instructional School for Teachers (IST) on Advanced Linear Algebra held at the Indian Institute of Technology Gandhinagar (IITGN), India: *July 10–22, 2017*.
8. Indian Academy of Sciences Summer Research Fellowship 2016 under the guidance of Prof. Jaydeb Sarkar at the Indian Statistical Institute (ISI), Bangalore, India: *May 2–June 27, 2016*.
9. Indian Institute of Science Education and Research (IISER) Mohali Summer Internship program under the guidance of Prof. I.B.S. Passi at IISER Mohali, India: *June 1–June 30, 2015*.
10. Mathematics Training and Talent Search Programme (MTTS, Level-O) at the Indian Institute of Technology Guwahati (IITG), India: *June 23–July 19, 2014*.

TECHNICAL SKILLS

- **Programming and Tools:** Python, Wolfram Mathematica, LaTeX, GitHub.
- **Teaching Platforms:** Canvas, Brightspace, Gradescope, Ed Discussion.